VTO1 – VOLUME TRANSFER AND NON-GAUSSIAN CONTINUUM COMPONENTS

Abstract. This focused outline follows the written proof of volume transfer and component cumulants in VTM1. It separates the normalization defect, the all-flux product estimate, the prescribed-parent moment passage and the general invariant-expectation lemma. The target is a positive fourth cumulant in the same continuum components used by the field-strength construction.

Contents

1. Introduction and target statement 1
Proof overview 3
2. D. The law, the probe and the imported bounds 4
3. VT1. Normalized transfer from two reference volumes 6
4. VT2. The photon and flux contributions to the defect 7
5. VT3. From zero-flux response to all-flux Wick products 8
6. T17. A positive charge exponent after volume transfer 9
7. T18. Moment convergence for the prescribed parent 10
8. B5. Distinct-point products bound invariant expectations 11

1. Introduction and target statement

VTO1 accompanies the 18-page VTM1 manuscript and expands the corresponding unit of QEO1. The full proof has difficulty 4/5. Each green box has a written proof at its stated scope; this records drafting, not independent certification. The opening states the component implication with its inputs explicit. The chamber-law definitions and reference estimates are recalled on D and T17.

Overall plan: QEO1/A3. The manuscript is VTM1, Theorem A.

A3. Non-Gaussianity in the same continuum components

Model-specific. Overview Target.

Fix a charge $0 < e \le 1$, a real $f \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$, and a translation-invariant law μ on tempered two-form distributions on $\mathbb{R}^3$. Let F_i be the physical step fields of D at cutoffs $N_i, T_i \to \infty$, with $\eta_i = L^{-N_i}$. Suppose their laws converge weakly to this prescribed μ on H_{-4}. Assume the moment and varying-test conclusions of T18. For real $f_i \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ with one common compact support, suppose $f_i \to f$ in L^2 and $v_i \ge 0$, $v_i \to v = \|f\|_2^2$. Define

$$H_2(x;v) = x^2 - v, \quad H_4(x;v) = x^4 - 6vx^2 + 3v^2, \qquad W_m(x;v,c) = e^{-m}H_m(x;v) - c.$$

Here e^{-m} is a power of the charge. Assume real bounded sequences $c_{2,i}, c_{4,i}$ and nonnegative $u_{2,i}, u_{4,i}$ such that, for some fixed $\lambda > 0$, every finite $Y \subset \lambda\mathbb{Z}^3$ obeys

$$\left| \mathbb{E}_i \prod_{y\in Y} W_m(F_i(\tau_y f_i); v_i, c_{m,i}) \right| \le u_{m,i}^{|Y|}, \qquad m = 2, 4,$$

for all sufficiently large i depending on Y. Suppose, with fixed $C < \infty$, $\sigma > 0$, $b(f) \in \mathbb{R}$ and $\eta_i \to 0$, that

$$u_{4,i} \le Ce^{\sigma}, \qquad (1 + |c_{2,i}|)u_{2,i} \le Ce^{\sigma},$$
$$|c_{4,i} - 3c_{2,i}^2 - b(f)| \le C(e^{\sigma} + \eta_i^{1/4}).$$

Finally, assume that almost every component in the translation-ergodic decomposition $\mu = \int \mu_\xi \, d\nu(\xi)$ satisfies

$$\mathbb{E}_{\mu_\xi} \exp(F(g)) \le \exp(\|g\|_2^2/2) \quad (g \in \mathcal{S}(\mathbb{R}^3; \Lambda^2) \text{ real}).$$

Then, on one ν-full set,

$$|\kappa_4^{\mu_\xi}(f) - e^4 b(f)| \le (8C + 3C^2)e^{4+\sigma}.$$

If $b(f) > 0$ and $(8C + 3C^2)e^{\sigma} \le b(f)/2$, these components have $\kappa_4^{\mu_\xi}(f) \ge e^4 b(f)/2 > 0$ and are non-Gaussian. The full-measure set is for the fixed charge, probe and prescribed parent. In the chamber construction, T17 supplies the product and coefficient bounds, with $C = C_U$. ECM1 supplies the component Gaussian bound; QEM1 checks the common parameter choice and the positive charge interval.

Why products are needed. A fourth-cumulant estimate for a mixture does not control its components. The product bound feeds the general conditional-expectation lemma B5. Extract a common convergent subsequence of $c_{2,i}, c_{4,i}, u_{2,i}, u_{4,i}$ without changing μ. After T18 passes the products to μ, apply B5 to $G_m = W_m(F(f); v, c_m)$. This gives $|d_m| \le u_m$ for $d_m = \mathbb{E}_{\mu_\xi} G_m$ on one full set. The component Gaussian bound gives zero mean, and

$$e^{-4}\kappa_4^{\mu_\xi}(f) = c_4 + d_4 - 3(c_2 + d_2)^2.$$

The error is bounded by $Ce^{\sigma} + u_4 + 6|c_2|u_2 + 3u_2^2$. Intersecting this component set with any structural full-measure set preserves the parent and its component laws.

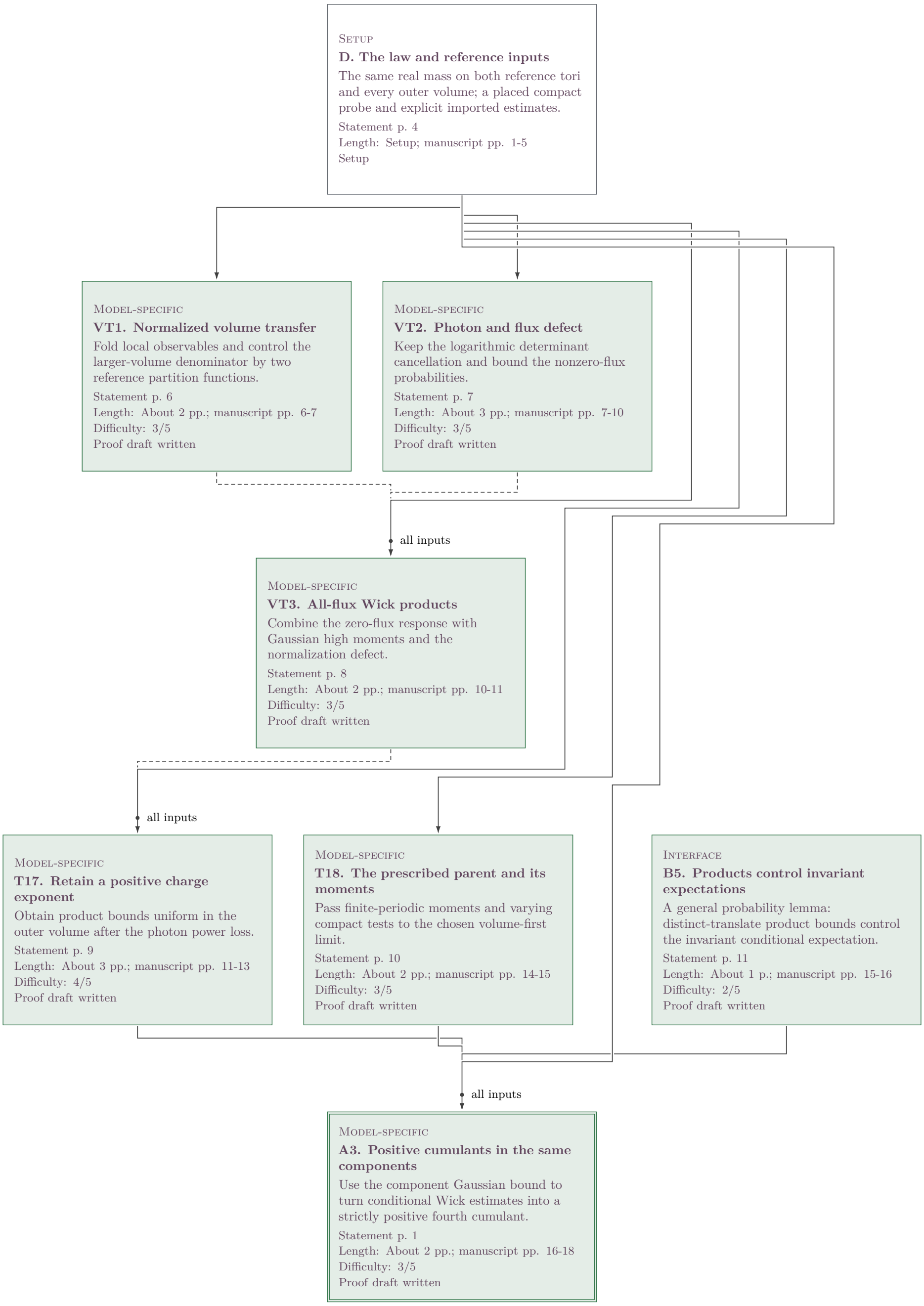
Source. VTM1, Proposition 8.2 and Theorem A; proof pp. 16–18. The chamber application uses the explicit hypotheses in D, T17 and T18.

Written proof: approximately 2 pages within VTM1 §8, sharing B5. Difficulty 3/5 for this passage; 4/5 for the full 18-page VTM1 manuscript. Page allocations below overlap at section boundaries and are not additive.

VTO1 – Volume transfer and non-Gaussian continuum components

Proof draft written | Partial coverage | Not drafted

Each box links to a complete statement page, with a further link to the current manuscript result. The component Gaussian bound is a separate input at A3. B5 is a standalone general lemma.



Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.

Click a box for its statement. Return links are navigation, not proof dependencies.

Lengths are approximate allocations in the current 18-page manuscript, with overlapping boundary pages. They are not additive budgets. Difficulty: 1 routine, 2 standard methods, 3 substantial, 4 hard specialist analysis, 5 several demanding mechanisms. Colors show proof-draft status; mathematical categories remain text labels.

2. D. The law, the probe and the imported bounds

Setup. Overview Target.

Finite laws. Use the chamber operator $D^{\mathrm{ch}}[B]$ with phase argument B, four spin components, 48 chamber labels, weights $(6, 6\sqrt{2}, 6\sqrt{2}, 6)$, $0 < \rho < 1$, and antiperiodic fermions. On a coordinate torus with V sites, set $r_q = 2\pi/q$ and

$$Z[O] = \sum_{z \in C^2(\mathbb{Z}),\ dz=0} \int_{[0,r_q)^{3V}} O(dA + r_q z) e^{-\|dA + r_q z\|^2/2} \det(D^{\mathrm{ch}}[qA] + \boldsymbol{\mu})\, dA.$$

The law is $Z[O]/Z[1]$. The sum retains all fluxes, with the identification $(A, z) \sim (A + r_q k, z - dk)$. Norms count positively oriented cells. LPM1 supplies positivity, coordinate slab traces, chessboard inequalities, translation and charge-conjugation invariance, and $\mathbb{E} e^{\langle g, F^{\mathrm{u}} \rangle} \le e^{\|g\|^2/2}$ for $F^{\mathrm{u}} = dA + r_q z$ and every real two-cochain g.

Scales and a common mass. Fix an odd integer $L \ge 13$, $M = L^k$ with integer $k \ge 2$, $0 < \bar{m} \le 1$, $0 < \varepsilon \le 1/200$, and set

$$h = L^{\lfloor \log_L(1/\bar{m}) \rfloor}, \quad \eta = L^{-N}, \quad q = e\sqrt{\eta}, \quad R = 8hL^T, \quad n = R/\eta,$$
$$v_K = -\log(e\sqrt{h}), \quad r_K = \min\{L^i : i \in \mathbb{Z}_{\ge 0},\ L^i \ge v_K^2\}, \quad R^* = hMr_K/L = hL^{T^*},$$
$$n_* = R^*/\eta, \quad N_* = 8L^3, \quad \ell_c = R^*/(2L), \quad a_L = (L^3 - 1)/(L - 1)^3.$$

The cutoffs N, T are nonnegative integers. At each e, N, every torus uses the same real lattice mass $\boldsymbol{\mu}_N$, independent of T. It may be signed. The step field $F^\sharp$ equals $\eta^{-3/2} F^{\mathrm{u}}$ on the corresponding plaquette cubes of side η.

The local probe. Fix real nonzero $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, set $f = dd^*\beta$, and let D_β be the side of a least support cube. Enlarge it by two in each direction to Q_β, with side $D_\beta^{\mathrm{buf}} = D_\beta + 4$ and center z_β. With J_η the cube average and $\delta_{\eta,\mu}$ the centered difference of step η, put

$$a'_{\eta,\nu} = -\sum_{\mu \ne \nu} \delta_{\eta,\mu} \beta_{\mu\nu}, \quad f^\eta_{\mu\nu} = \delta_{\eta,\mu} a'_{\eta,\nu} - \delta_{\eta,\nu} a'_{\eta,\mu},$$

and $\widehat{a}^o_\nu(x) = \eta^{1/2} (J_\eta a'_{\eta,\nu})(\eta x + \eta e_\nu/2)$. Write $v_\eta = \|d\widehat{a}^o\|^2$. The periodized, oriented reflected copies in $\{0, \dots, 2L-1\}^3$ are $\widehat{a}_c$. Their centered placement $v \in \eta\mathbb{Z}^3$ satisfies

$$|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \le \eta/2, \quad D_\beta^{\mathrm{buf}} + 3\eta \le \ell_c, \quad v + Q_\beta \subset [\ell_c/8, 7\ell_c/8]^3.$$

RRM1 supplies $d\widehat{a}^o = (f^\eta)_{[\eta]}$, pairwise orthogonal curvatures of variance v_η, common compact support, $f^\eta \to f$ in L^2, and $v_\eta \to \|f\|_2^2$.

Reference quantities. On side n, let γ_n be standard Gaussian measure on exact two-cochains, $A(u)$ their coclosed mean-zero potential and $H_{\theta,\nu} = \theta_\nu/(qn)$. The positive matter factor is

$$z_{r,n} = \int \frac{d\theta}{(2\pi)^3} \int \det(D^{\mathrm{ch}}[q(A(u) + H_\theta)] + \boldsymbol{\mu}_N)\, d\gamma_n(u), \quad \theta \in [-\pi, \pi]^3.$$

Use the ordinary derivatives $c_m = N_0^{(m)}(0)$, $m = 2, 4$, of

$$N_0(t) = \frac{\det(D^{\mathrm{ch}}[q(t/e)\widehat{a}_0] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)}.$$

This free denominator has positive mass. With W_m defined on A3, $X_c = \langle d\widehat{a}_c, F^{\mathrm{u}} \rangle$, and expectation $\mathbb{E}^{(0)}$ conditioned on zero flux, define

$$\Theta_m^{(0)} = \left| \mathbb{E}^{(0)}_{n_*} \prod_c W_m(X_c; v_\eta, c_m) \right|^{1/N_*}.$$

The source identity identifying this product with the differentiated normalized response is RRM1 Proposition 3.2. The continuum coefficient is

$$b(f) = -6\,\mathrm{tr}\left[((\bar{m} + \gamma \cdot \partial)^{-1} i\gamma \cdot d^{*}\beta)^{4}\right].$$

PRM1 gives a probe with $b(f) > 0$; positivity is needed only at A3. TPM1 supplies the matter comparison and RRM1 the response bound listed explicitly on T17. CEM1 supplies prescribed volume-first laws; ECM1 supplies Gaussian domination on their components. These inputs have separate roles and are not consequences of volume transfer.

Source. VTM1 §2, Theorem A and its introduction.

Setup and imports: VTM1 pp. 1–5. No independent proof-length allocation.

3. VT1. Normalized transfer from two reference volumes

Model-specific. Overview Target.

Use the finite chamber laws of D, with fixed charge $q > 0$ and real mass. Choose cells of width $w \in \frac{1}{2}\mathbb{Z}_{>0}$ whose walls are admissible coordinate mirrors. Let $s_0 \geq 2$ be even, $n_r = ws_0 \geq 4$, $N_r = s_0^3$, and $j \geq 2$ an integer. For O in the scalar class $\mathcal{A}(Q_0^\circ)$ defined below, write $O^{[c]}$ for its reflected copy, including the antilinear action on coefficients. Require the support and its one-unit enlargement to embed as in the slab observable class. Define

$$\|O\|_* = \left(\mathbb{E}_{C(n_r)} \prod_{c \bmod s_0} O^{[c]} \right)^{1/N_r}, \quad D_j = j^3 \log Z_{n_r}[1] - \log Z_{jn_r}[1], \quad \delta_j = \frac{D_j}{(j-1)^3 N_r}.$$

The fold trace makes the quantity in the root nonnegative. Here $\mathcal{A}(U)$ is the linear span of $\Phi(d\widehat{A}) \prod_j G_j(\widehat{A}|_{B_j})$, with Φ bounded Borel in finitely many curvature values and pairwise disjoint coordinate boxes $B_j \Subset U$. Each G_j is Borel and gauge invariant, and for some $C_j < \infty$, $\epsilon_j > 0$ satisfies

$$|G_j(\widehat{A})| \leq C_j \exp\left[(\tfrac{1}{2} - \epsilon_j) \sum_{\sigma \in B_j^{(2)}} |d\widehat{A}(\sigma)|^2 \right].$$

The real local lift obeys $d\widehat{A} = F^{\mathrm{u}}$ and $\exp(iq\widehat{A}) = \exp(iqA)$; gauge invariance makes its evaluation independent of the lift. Each enlarged box and its period translates must be disjoint, and distinct boxes must remain disjoint under periods. Then $D_j \geq 0$. If s is even and either $s = s_0$ or $s \geq js_0$, on $C(ws)$ every finite set $\mathcal{C}$ of distinct cells obeys

$$\left| \mathbb{E}_{C(ws)} \prod_{c \in \mathcal{C}} O_c^{[c]} \right| \leq \prod_{c \in \mathcal{C}} \exp(\delta_j) \|O_c\|_*.$$

The normalization argument. The slab identity is $\mathcal{Z}_s(O) = \mathrm{Tr}(Y^* Y)^{s_\mu/2}$. Schatten monotonicity bounds the folded numerator by its reference value. For the denominator, convexity of $\log Z$ in each side gives, with $r = s/s_0 \geq j$,

$$\log Z_{ws}[1] \geq r^3 \left(\log Z_{n_r}[1] - \frac{D_j}{(j-1)^3} \right).$$

The three negative terms in the successive convexity estimates are bounded using side monotonicity. Combine this lower bound with the rectangular chessboard inequality. No thermodynamic pressure limit is used.

Source. VTM1 Proposition 3.1, pp. 6–7.

Proof allocation: about 2 pages; difficulty 3/5.

4. VT2. The photon and flux contributions to the defect

Model-specific. Overview Target.

For the cubic all-flux law in D, fix the same $q > 0$ and real mass on sides n and jn, where $n \geq 4$ and $j \geq 2$ is an integer. Let $P_n(w)$ be the probability of flux $w \in \mathbb{Z}^3$ and

$$\varsigma_n = \sum_{w \neq 0} \exp[-2\pi^2 |w|^2/(q^2 n)].$$

Then $P_n(w) \leq \exp[-2\pi^2|w|^2/(q^2n)]$ and $P_n(0) \geq 1 - \varsigma_n$. If $q^2 n \leq 1$, $\varsigma_n \leq 7\exp[-2\pi^2/(q^2n)] < 1/8$. The defect $D_j = j^3 \log Z_n[1] - \log Z_{jn}[1]$ decomposes as

$$\begin{aligned} D_j &= D_j^{\mathrm{ph}} + D_j^{\mathrm{mat}} + D_j^{\mathrm{fl}}, \\ D_j^{\mathrm{mat}} &= j^3 \log z_{r,n} - \log z_{r,jn}, \\ D_j^{\mathrm{ph}} &\leq (j^3 - 1) \log \frac{2\pi}{q^2 n} + K_j, \\ D_j^{\mathrm{fl}} &\leq -j^3 \log(1 - \varsigma_n) \quad (\varsigma_n < 1), \end{aligned}$$

where $K_j = (1 + j + j^2)(\log j + jG_j)$ and $G_j = (1 + j/3.2)2j^2/2.56$.
What must be retained. Let $\mathcal{L}(n)$ denote the logarithmic determinant of the cubic scalar lattice Laplacian with its constant mode omitted. The needed estimate is

$$\tfrac{1}{2}[\mathcal{L}(jn) - j^3 \mathcal{L}(n)] \leq -(j^3 - 1)\log n + K_j.$$

Summing one Fourier coordinate gives a hyperbolic-sine product; telescoping the three side enlargements preserves the displayed logarithmic term. The exact zero-flux Gaussian normalization then gives the photon bound. Gaussian domination tested on constant harmonic two-cochains gives the sector probabilities. The matter term remains an independent TPM1 input.

Source. VTM1 §4, especially Proposition 4.1, pp. 7–10.

Proof allocation: about 3 pages; difficulty 3/5.

5. VT3. From zero-flux response to all-flux Wick products

Model-specific. Overview Target.

Use the probe and reference array in D, with $0 < e \le 1/3$ and

$$6N_* e^2 R^* \log(1/e) \le \pi^2.$$

Set $D^{\mathrm{mat}} = L^3 \log z_{r,n_*} - \log z_{r,Ln_*}$, $\delta = D_L/((L-1)^3 N_*)$, and

$$M_4 = 600 N_*^2 v_\eta^2 e^{-4} + |c_4|, \quad M_2 = 12 N_* v_\eta e^{-2} + |c_2|,$$

$$u_m = \exp(\delta)(\Theta_m^{(0)} + 2e^6 M_m).$$

For $n = n_*$, or $n \in (n_*/L)\mathbb{Z}$ with $n \ge Ln_*$, every set $\mathcal{C}$ of distinct cells satisfies

$$\left| \mathbb{E}_n^{\mathrm{af}} \prod_{c \in \mathcal{C}} W_m(X_c; v_\eta, c_m) \right| \le u_m^{|\mathcal{C}|}, \quad m = 2, 4,$$

and

$$0 \le \delta \le \frac{a_L}{N_*} \log \frac{2\pi}{e^2 R^*} + \frac{K_L + D^{\mathrm{mat}} + 1}{(L-1)^3 N_*}.$$

The flux error. The fully folded product on the reference torus is nonnegative. Its zero-flux part is bounded by $(\Theta_m^{(0)})^{N_*}$. Gaussian high-moment bounds give $\|W_m(X_c)\|_{2N_*} \le M_m$; Cauchy–Schwarz and VT2 bound the other sectors by $(2e^6 M_m)^{N_*}$. Apply VT1 to the resulting reference root. The one-cell case also yields

$$|e^{-4}\kappa_4 - (c_4 - 3c_2^2)| \le u_4 + 3u_2(2|c_2| + u_2).$$

This last inequality concerns the finite law; the component conclusion still requires the full family of product bounds.

Source. VTM1 Proposition 5.1, pp. 10–11.

Proof allocation: about 2 pages; difficulty 3/5.

6. T17. A positive charge exponent after volume transfer

Model-specific. Overview Target.

Use D and its support placement. In addition require $v_K \geq 8$, $e \leq 1/3$, $e\sqrt{h}\log h \leq 1$, and

$$6N_* M v_K^2 \exp(-v_K)(v_K \exp(-v_K) + \tfrac{1}{2}) \leq \pi^2.$$

For all charges in the admitted interval and $N \geq N_{\mathrm{ref}}(e)$, assume constants $C_{\mathrm{mat}}, C_{\mathrm{ref}} < \infty$ independent of charge, mesh and placement such that

$$\begin{aligned} D^{\mathrm{mat}} &\leq C_{\mathrm{mat}}, \\ \Theta_4^{(0)} \leq C_{\mathrm{ref}} e^{\mathfrak{e}}, &\quad (1+|c_2|)\Theta_2^{(0)} \leq C_{\mathrm{ref}} e^{\mathfrak{e}}, \\ |c_2| + |c_4|^{1/2} \leq C_{\mathrm{ref}}, &\quad |c_4 - 3c_2^2 - b(f)| \leq C_{\mathrm{ref}}(e + \eta^{1/4}), \end{aligned}$$

where $\mathfrak{e} = \frac{1}{2}(1/4 - 9\varepsilon) - 12\varepsilon$ and $\sigma = \mathfrak{e} - 2a_L/N_* > 0$. Then a constant $C_U < \infty$, independent of e, N, T and placement, gives the bounds

$$u_4 \leq C_U e^{\sigma}, \quad (1+|c_2|)u_2 \leq C_U e^{\sigma}, \quad |c_4 - 3c_2^2 - b(f)| \leq C_U(e^{\sigma} + \eta^{1/4}).$$

For $T \geq T^* + 1$ and finite $Y \subset (R^*/L)\mathbb{Z}^3$ of points incongruent modulo $R\mathbb{Z}^3$,

$$\left| \mathbb{E}_{N,T} \prod_{y \in Y} W_m(F^\sharp(\tau_y f^\eta); v_\eta, c_m) \right| \leq u_m^{|Y|}, \quad m = 2, 4.$$

The smooth-test fourth cumulant satisfies

$$|\kappa_4(F^\sharp(f)) - e^4 b(f)| \leq (8C_U + 3C_U^2)e^{4+\sigma} + C_U(\eta^2 + e^4\eta^{1/4}).$$

The loss that matters. The photon term gives $\exp(\delta) \leq C_\Gamma e^{-2a_L/N_*}$. Multiply this by the reference error $e^{\mathfrak{e}}$ in VT3. The strict positivity of σ is the required margin. Even cell indices represent translations on $(R^*/L)\mathbb{Z}^3$; their distinctness is exactly the incongruence condition above. Gaussian multilinear moment estimates control the replacement of f^η by f.

Source. VTM1 Proposition 6.1 and its proof, pp. 11–13. TPM1, RRM1 and PRM1 supply the three displayed reference inputs; QEM1 admits them on one common charge interval.

Proof allocation: about 3 pages; difficulty 4/5.

7. T18. Moment convergence for the prescribed parent

Model-specific. Overview Target.

Let P_{N_i,T_i} be the laws of the periodic step fields of D, with $N_i, T_i \to \infty$, and suppose $P_{N_i,T_i} \Rightarrow \mu$ on

$$H_{-4} = \{F : \sum_{\kappa} \lambda_\kappa^{-8} |F(h_\kappa)|^2 < \infty\}.$$

Here h_κ are real Hermite two-forms and λ_κ the eigenvalues of $1 + |x|^2 - \Delta$. Then $\mathbb{E}_\mu e^{F(g)} \le e^{\|g\|_2^2/2}$ for every real Schwartz two-form g, and every joint moment of a fixed finite family of Schwartz tests converges along that same sequence.

The finite-volume bound. For physical period R, write $g^R = \sum_{k\in\mathbb{Z}^3} g(\cdot + Rk)$. Periodization and cell averaging give

$$\mathbb{E}_{N,T} e^{F^\sharp(g)} \le e^{p_R(g)^2/2}, \quad p_R(g) = \|g\|_2 + 26R^{-2} \||x|^2 g\|_{L^2(\mathbb{R}^3 \setminus [-R/2,R/2)^3)}.$$

For $R \ge 1$, $p_R \le p_1$ and $p_R(g) \to \|g\|_2$. The uniform exponential bound makes the generating functions locally uniformly bounded; uniform integrability and Cauchy's formula pass their derivatives to the limit. If the supports lie in one open period cube, a degree-J monomial changes by at most $Jc_J B^{J-1}\delta$ when each test changes by δ in L^2 and all test norms are at most B.

Preserving the chosen law. Suppose instead that μ_i are prescribed fixed-spacing volume limits, with $N_i \to \infty$ and $\mu_i \Rightarrow \mu$. Choose a torus from each defining volume sequence with weak distance less than $1/i$ from μ_i and $T_i \ge i + T^* + 1$. These finite-torus laws still converge to μ. A further subsequence making the bounded reference coefficients converge also preserves μ. Each fixed finite family of translated compact tests eventually fits and has distinct torus translates. This result supplies moment passage, not the separate component Gaussian bound.

Source. VTM1 Proposition 7.1 and §7, pp. 14–15.

Proof allocation: about 2 pages; difficulty 3/5.

8. B5. Distinct-point products bound invariant expectations

Interface. Overview Target.

Let $d \geq 1$ be an integer. Let $U(y)$, $y \in \mathbb{Z}^d$, arise from a probability-preserving $\mathbb{Z}^d$ action, and let P_{lat} be conditional expectation onto its invariant sigma-algebra. Suppose G is real and belongs to every finite L^p. If $u \geq 0$ and

$$\left| \mathbb{E} \prod_{a=1}^{J} U(y_a) G \right| \leq u^J$$

for every finite family of pairwise distinct points $y_a \in \mathbb{Z}^d$, then $|P_{\mathrm{lat}} G| \leq u$ almost surely. If this is the restriction of a probability-preserving $\mathbb{R}^d$ action to $\lambda\mathbb{Z}^d$, its full invariant conditional expectation P also satisfies $|PG| \leq u$. When the full action has a conditional-law ergodic decomposition $\mu = \int \mu_\xi \, d\nu(\xi)$, $|\mu_\xi(G)| \leq u$ for almost every component.

Proof mechanism. Average G over a cube of n^d lattice points. In its Jth moment for even J, the distinct-index terms obey the assumed bound. There are at most $\binom{J}{2} n^{d(J-1)}$ repeated-index terms, controlled by $\|G\|_J^J$. The mean ergodic limit and Fatou's lemma give $\mathbb{E}|P_{\mathrm{lat}} G|^J \leq u^J$. Let $J \to \infty$. For the full translation action, $P = P P_{\mathrm{lat}}$ and positivity of conditional expectation gives the last assertion. No lattice-ergodicity assumption is required.

Source. VTM1 Lemma 8.1 and the paragraph following it, pp. 15–16.

Proof allocation: about 1 page; difficulty 2/5. This is a general interface.